LOCAL LLM / CUSTOMER ANALYTICS

LLM Customer Segmentation Assistant

Explore customer churn data in plain English with clarification questions, inspectable SQL and local execution.

Natural language

Find customer segments without writing SQL manually.

Clarify first

Ask for missing criteria before querying customer data.

Traceable results

Inspect generated SQL and download the result as CSV.

FROM A QUESTION TO A CUSTOMER SEGMENT

The business need

Turn churn scores and customer attributes into targeted customer lists and group comparisons.

How the assistant works

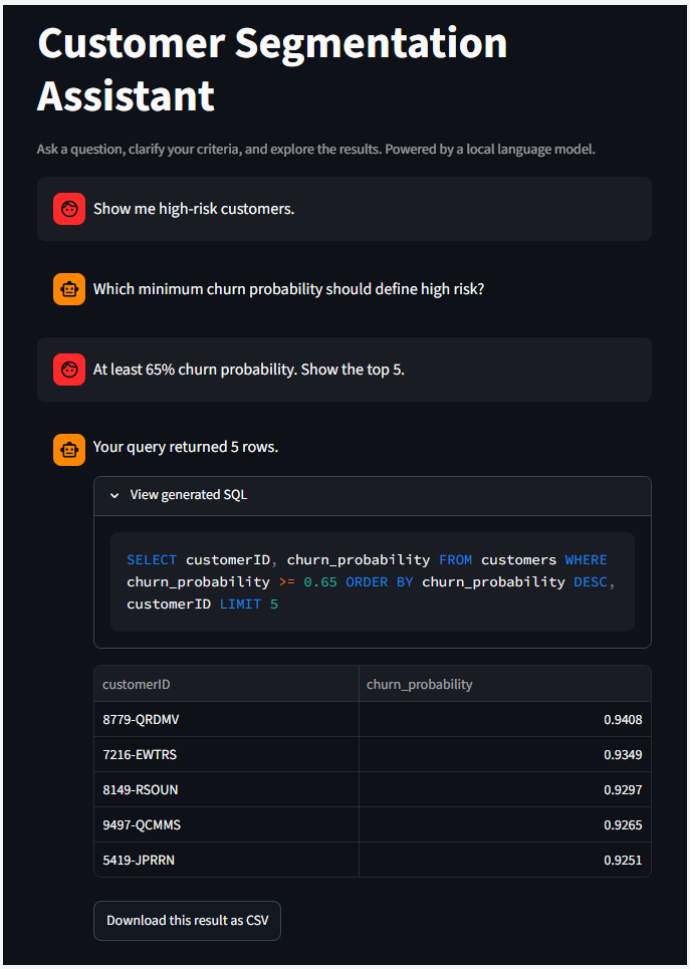
A local model translates the request into a structured decision. DuckDB computes the result after SQL validation.

Local and controlled

Read-only access to one customer table. Unsupported requests receive an explanation; CSV downloads require a user action.

APPLICABLE PATTERN

Natural-language inputs connected to deterministic data queries and calculations. The same separation can support other Python analysis workflows.



Actual application: an undefined risk category is clarified before the query runs.

DEVELOPMENT VALIDATION

6,734 customers | 9/9 reference cases passed

Reference cases cover filters, aggregates, rankings and a clarification dialogue. Several informed prompt development; this is not an independent accuracy benchmark.

Python · Ollama · Qwen2.5 Coder 3B · DuckDB · SQLGlot · Streamlit

Independent portfolio prototype using public Telco data and upstream model scores. No observed churn outcomes, public deployment or production-readiness claim.